

PROVISIONAL DATA SHEET

DEVELOPMENT PRODUCT

Grilamid FE 11859 black 9675

Product description:

Grilamid FE 11859 black 9675 is a specially developed thermoplastic polyamide showing a combination of high ductility and high impact strength. It is designed for injection molding processes and suitable for thin walled parts.

Key features are:

- High surface hardness and scratch resistance
- High stress crack resistance

Processing:

Processing conditions for injection molding:

Following parameters are recommended for processing of Grilamid FE 11859 black 9675

Melt temperature:	280 - 320 °C
Mold temperature:	70 - 100 °C
Injection speed:	medium – high
Holding pressure (spec.):	> 600 bar
Dynamic pressure (spec.):	50 - 150 bar
Peripheral screw speed:	0.05 - 0.3 m/s

The material is delivered dry and ready for processing in sealed bags. If the granules in opened bag absorb moisture, the material can be dried in desiccant air dryers at 80°C for 4 to 10 h, depending on the moisture content, or preferably in a vacuum oven at 70°C up to 4 - 8 h. To ensure impeccable color during long storing periods, storage in heated environment should be avoided.

PROPERTIES

Mechanical Properties

		Standard	Unit	State	Grilamid FE 11859 black 9675
Tensile E-Modulus	50 mm/min	ISO 527	MPa	Cond.	2'300
Tensile strength at yield	50 mm/min	ISO 527	MPa	Cond.	65
Elongation at yield	50 mm/min	ISO 527	%	Cond.	5.0
Tensile strength at break	50 mm/min	ISO 527	MPa	Cond.	60
Elongation at break	50 mm/min	ISO 527	%	Cond.	> 100
Impact strength	Charpy, 23 °C	ISO 179/1eU	kJ/m ²	Cond.	No break
Notched Impact strength	Charpy, 23 °C	ISO 179/1eA	kJ/m ²	Cond.	16

Thermal Properties

Melting Point	DSC	ISO 11357	°C	dry	260
Glass transition temperature	DSC	ISO 11357	°C	dry	80
HDT/B	0.45 MPa	ISO 75	°C	dry	103
Thermal expansion coefficient long.	23-55°C	ISO 11359	10 ⁻⁶ /K	dry	75
Thermal expansion coefficient trans.	23-55°C	ISO 11359	10 ⁻⁶ /K	dry	80

General

Density		ISO 1183	g/cm ³	dry	1.14
Moisture absorption	23°C/50% r.F.	ISO 62	%		2.5
Linear mould shrinkage	long.	ISO 294	%	dry	0.40
Linear mould shrinkage	transverse	ISO 294	%	dry	0.45

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- Grilamid FE 11859 black 9675 is a development product.
- The listed properties are provisional and not committed.
- The commercialization of this product is not decided yet.